

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0627

North Madison 345 138 kV
September 29, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0627. The Interconnection Customer proposes a 301.2 MW solar photovoltaic facility interconnecting at the North Madison 345 138 kV substation in the West Virginia area. The request was received on June 08, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 278 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,570,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GLEN LAKE – DORRIS 138 KV	AMOS – COULEEAVE 345 KV Line	166.6	178	7.4
BIG FISH – TEAL 138 KV	COULEE AVE – AMOS 345 KV	139.9	222	6.2
MATTOON EAST – ROCKAWAY BEACH 138 KV	COULEE AVE – SPORN 345 KV	134.5	255	6.1
HAMPSHIRE PAPER – RCKAWAY BEACH 138 KV	COULEE AVE – SPORN 345 KV	133.6	255	6.1
WICCOPEE – TACKETT CRK SUBSTATION 138 KV	HURRICANE – N MADISO 345 138 KV	123.7	234	73.8
WICCOPEE – NORTH MADISON 345 138 KV line	HURRIANE – NORTH MADSON 345 138 KV	123.7	240	73.8
NORRIS STREET – RED MAPLE 138 KV	HURRICANE – NORTH MADISON 345 138 KV	120.7	162	21.7
HURRICANE – NORTH MADISON 345 138 KV	N MADISON 345 – WICCOPEE 138 KV	115.1	279	77.5
UNION VLY – POTOMAC YARD 138 KV	MATTOON E – ROCKAWAY BEACH 138 KV	114.9	251	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GLEN LAKE – DORRIS 138 KV	Rebuild line with higher-capacity conductor	\$36,414,000	14% (\$4,981,000)
TEAL – BIG FISH 138 KV	Rebuild line with higher-capacity conductor	\$11,764,000	11% (\$1,301,000)
MATTOON E – ROCKAWAY BEACH SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$10,213,000	13% (\$1,317,000)
ROCKAWAY BEACH – HAMPSHIRE PAPER 138 KV	Rebuild line with higher-capacity conductor	\$12,681,000	13% (\$1,636,000)
WICCOPEE – TACKETT CRK SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$6,546,000	75% (\$4,925,000)
WICCOPEE – NORTH MADISON 345 138 KV line	Rebuild line with higher-capacity conductor	\$2,864,000	75% (\$2,155,000)
NORRIS STREET – RED MAPLE 138 KV	Rebuild line with higher-capacity conductor	\$4,781,000	31% (\$1,461,000)
HURRICANE – NORTH MADISON 345 138 KV	Rebuild line with higher-capacity conductor	\$2,937,000	60% (\$1,756,000)
UNION VLY – POTOMAC YARD 138 KV	Rebuild line with higher-capacity conductor	\$7,860,000	13% (\$1,038,000)
	Total allocated to project		\$20,570,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.